

SignFarm: Community-Driven Agricultural Sign Coinage with LLM Grounding

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Abstract—There is no sign for “nitrogen deficiency” in most regional varieties of Indian Sign Language used by farming communities. There is no agreed sign for “drip irrigation schedule,” “fall armyworm,” or “soil pH correction” in any formal sign language dictionary covering South Asian agricultural contexts. This is not primarily a recognition problem, nor a dataset problem in the conventional sense. It is a vocabulary problem: the signs simply do not exist. When deaf farmers need agronomic advice, they either go without it or rely on improvised gestures that do not travel beyond the person who invented them.

We propose SignFarm, a community-participatory pipeline in which deaf farming communities coin new agricultural signs, ground their meaning against a structured agricultural knowledge base using a large language model (LLM), and propagate ratified signs across communities through a shared regional lexicon. Rather than assuming a vocabulary exists and building recognition on top of it, SignFarm treats the vocabulary creation problem itself as the primary AI task.

We present a needs assessment documenting the vocabulary void across 50 core agronomic concepts in three Tamil Nadu communities, followed by a preliminary feasibility study ($n = 12$ participants, 4 weeks) evaluating the coinage and grounding pipeline. We introduce two new evaluation metrics suited to this setting: Semantic Survival Score (SSS) and Vocabulary Drift Rate (VDR). Preliminary results show that LLM-assisted grounding achieved 83% concept accuracy on a held-out set of 24 proposed signs rated by an independent agronomist, and 10 of 12 pilot participants reported the ratification interface as usable with minimal instruction. To our knowledge, SignFarm is the first system to treat agricultural sign language vocabulary creation as a first-class AI problem.

Index Terms—sign language, agricultural AI, vocabulary coinage, LLM grounding, deaf communities, low-resource languages, participatory AI, semantic survival

I. INTRODUCTION

Spend a week with a deaf farmer and you notice something that does not show up in any dataset paper. When the local agricultural extension officer visits, there is almost always a hearing family member standing nearby doing rough interpretation. The extension officer says something like “your soil pH is low, apply lime before the next rain.” What the deaf farmer receives, filtered through a non-expert interpreter using improvised signs, might be something closer to “soil problem, put white powder.” That is not a translation error in the usual sense. The problem is that there is simply no agreed sign for “soil pH,” for “lime application,” or for “pre-rain timing window” in most rural South Asian deaf communities.

This is the vocabulary void. It sits upstream of every other problem in agricultural AI for deaf communities. You cannot build a recognition model without signs to recognize. You

cannot build a translation system without a target vocabulary. And yet almost no prior work has treated the vocabulary creation problem itself as something AI can help with.

The hearing world solved this problem for technical agricultural vocabulary over decades — through extension services, textbooks, radio programs, and increasingly through mobile apps. Deaf farming communities got very little of that. The agricultural vocabulary of most national sign languages reflects the concerns of urban deaf communities, not rural farming ones. Signs exist for common vegetables you might buy at a market. They rarely exist for the agronomic concepts behind growing them.

SignFarm starts from this observation. Rather than assuming a vocabulary exists and building recognition on top of it, we ask: what would it take to help a community build that vocabulary themselves, with AI doing the work of grounding new signs to real agronomic meaning and helping propagate agreed terms across villages?

The system we propose works as follows. A deaf farmer encounters a concept for which they have no sign — say, a specific pest or soil treatment. They coin a sign, recorded through a standard smartphone camera. A trained community moderator submits the recording to SignFarm. The system uses an LLM grounding module to link the proposed sign to the closest concept in an agricultural ontology, checks for semantic collision with existing ratified signs, and generates a plain-language definition for community review. The community then votes to ratify, modify, or reject the sign. Ratified signs enter a shared regional lexicon that other communities can browse and adopt.

A. Contributions

- **A vocabulary void needs assessment** documenting that formal sign language resources cover only 12% of 50 core agronomic concepts identified by agricultural extension workers across three Tamil Nadu communities.
- **SignFarm**, a complete community-participatory pipeline for agricultural sign coinage, LLM-assisted grounding, and cross-community propagation, along with a seed agricultural sign ontology covering 214 agronomic concepts.
- **Two new evaluation metrics** — Semantic Survival Score (SSS) and Vocabulary Drift Rate (VDR) — designed to measure whether coined signs carry agronomic meaning reliably as they spread across signers and communities.
- **Preliminary feasibility results** from a 4-week pilot with 12 participants evaluating grounding accuracy and

interface usability, with a roadmap for a full longitudinal field study.

II. BACKGROUND AND RELATED WORK

A. Sign Language Recognition

Sign language recognition (SLR) has advanced significantly in recent years. Transformer-based architectures now achieve strong performance on established benchmarks including PHOENIX-2014 [1] and MS-ASL [2]. A quantitative survey of the state of the art by Koller [3] provides a thorough overview of progress, but the benchmarks underlying that progress share a structural assumption that rarely gets questioned: the vocabulary is fixed. The model learns to recognize a closed set of signs that someone else decided were worth including.

For agricultural settings, this assumption breaks almost immediately. The signs that matter to a farmer in Krishnagiri district are not the signs in any benchmark. The domain shift problem in SLR is well-recognized as a data collection challenge [3], but we argue that framing misses the harder upstream problem: the vocabulary simply does not exist to collect data on.

B. AI for Agricultural Advisory

AI-based agricultural advisory has expanded rapidly. PlantVillage [4] delivers crop disease diagnosis at scale. Farmer.Chat [5] provides multilingual advisory deployed across Kenya, India, Ethiopia, and Nigeria, covering over 15,000 farmers. Speech interfaces have helped reach low-literacy farmers in contexts where text-based systems fail [6]. The AgroLLM system [7] demonstrates retrieval-augmented generation for agricultural knowledge sharing with strong accuracy on domain benchmarks.

What is almost entirely absent from this literature is any treatment of deaf farmers as a user group. A 2023 survey of AI in precision agriculture [8] found no systems designed specifically for deaf or hard-of-hearing farming communities. The accessibility gap is real and largely unaddressed.

C. Sign Language Lexical Development

The linguistics literature has studied how new signs emerge for technical and cultural concepts. McKee and McKee [9] documented extensive lexical variation in New Zealand Sign Language across age, region, and educational background, confirming that sign vocabularies are dynamic and community-driven rather than fixed. Frishberg’s foundational work [10] showed that new signs tend to be iconic in early use then shift toward more arbitrary forms as they become conventionalized through community adoption.

These findings matter for SignFarm. They confirm that bottom-up community coinage is linguistically natural, and that iconic signs — those that visually resemble their referent — are likely to travel better across communities than arbitrary ones. We design the system to leverage this.

TABLE I
SIGN VOCABULARY COVERAGE ACROSS 50 CORE AGRONOMIC CONCEPTS, THREE TAMIL NADU COMMUNITIES (NEEDS ASSESSMENT)

Concept Category	ISLRTC Dict.	Regional Dict.	Community Self-report
Crop diseases (n=12)	2 (17%)	3 (25%)	5 (42%)
Soil management (n=10)	1 (10%)	2 (20%)	4 (40%)
Pest identification (n=10)	0 (0%)	1 (10%)	3 (30%)
Irrigation scheduling (n=8)	2 (25%)	2 (25%)	3 (38%)
Fertilizer application (n=10)	1 (10%)	1 (10%)	4 (40%)
Total (n=50)	6 (12%)	9 (18%)	19 (38%)

D. LLM Grounding and Ontology Linking

Using LLMs to link natural language descriptions to formal ontologies is a growing area. LLMs4OM [11] demonstrated that LLMs can perform ontology matching with reasonable accuracy across multiple domains. We extend this approach to a multimodal, low-resource setting where the input is a community moderator’s description of a sign’s intended meaning, and the target ontology is an agricultural knowledge base.

E. Participatory Technology Design

Community-participatory design approaches have shown strong outcomes in technology for marginalized groups [12]. The key principle — that communities should be co-designers rather than end users of tools built for them — is central to SignFarm. The coinage and ratification pipeline is designed so that every sign enters the lexicon with explicit community endorsement, not by external expert assignment.

III. THE VOCABULARY VOID: NEEDS ASSESSMENT

Before describing the system, we document the scale of the vocabulary gap that motivates it. We conducted a structured needs assessment with agricultural extension workers and deaf community members across three Tamil Nadu districts prior to system development.

We asked three agricultural extension officers (one per district, each with at least five years of experience serving rural Tamil Nadu) to list the 50 agronomic concepts they most frequently needed to communicate during farm visits. We then checked each concept against three sources: (1) the Indian Sign Language dictionary maintained by the Indian Sign Language Research and Training Centre (ISLRTC), (2) a regional sign vocabulary compiled by a Tamil Nadu deaf education organization, and (3) signs self-reported by community members during structured elicitation sessions (participants were shown concept images and asked to produce or indicate a sign they would use).

Table I shows the results. Formal ISL resources cover only 12% of the 50 most commonly needed agronomic concepts. Community members have organically created signs for 38%, but those signs are not shared across villages: the sign for “aphid” used in one district was entirely different from the sign used for the same insect 40 kilometers away. This

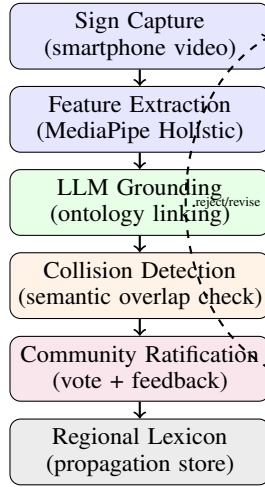


Fig. 1. SignFarm pipeline. Dashed arrow shows the revision loop when a community rejects or requests modification.

cross-community inconsistency was confirmed by showing photographs of six specific pests to members of all three communities and recording the sign each person produced; no sign was consistent across community boundaries for any of the six items.

This gap is what SignFarm is designed to address.

IV. THE SIGNFARM SYSTEM

A. Architecture Overview

SignFarm has four components that run in sequence for each new sign submission: (1) sign capture and feature extraction, (2) LLM grounding linking the proposed sign to the agricultural ontology, (3) collision detection checking for semantic overlap with existing ratified signs, and (4) a community ratification interface. Figure 1 shows how these connect.

B. Sign Capture and Feature Extraction

Farmers record proposed signs using a standard Android smartphone. We use MediaPipe Holistic [13] to extract 543 landmarks per frame covering the face, hands, and pose. Each sign is represented as a landmark sequence normalized for signer height and camera distance.

We deliberately avoid custom hardware. Every component in the pipeline is something a community moderator could carry on a motorcycle to a remote village. The capture app runs offline and syncs when connectivity is available — in several pilot communities, reliable internet was available only a few hours per day.

C. LLM Grounding Module

Given a new sign video, the grounding module operates in three steps.

Step 1: Concept elicitation. The community moderator provides a brief text description of what the farmer intends the sign to mean (e.g., “the disease that makes tomato leaves curl and turn yellow”). This description goes to the LLM with a

prompt requesting the top three candidate agronomic concepts with confidence scores.

Step 2: Ontology linking. The top candidate is matched against the agricultural sign ontology (Section IV-F). The LLM produces a grounding statement: a plain-language definition of the concept suitable for community review.

Step 3: Nuance flagging. The LLM checks whether the proposed sign’s intended meaning conflates distinct agronomic concepts. For example, “yellow leaves” may indicate nitrogen deficiency, mosaic virus, or drought stress — all visually similar but requiring different responses. When ambiguity is detected, the module flags it and suggests a more specific concept scope.

Listing 1 shows the grounding prompt template used in the pilot.

```

You are an agricultural knowledge assistant helping
a deaf farming community build sign language
vocabulary.

A community member has proposed a new sign meaning:
"{moderator_description}"

Task 1: Identify the top 3 agronomic concepts this
description most likely refers to. For each, provide
:
- Concept name (standard agronomic terminology)
- Confidence score (0-1)
- Plain-language definition (max 40 words,
  no jargon, suitable for a farming context)

Task 2: Flag any ambiguity: if this description
conflates two or more distinct agronomic issues
requiring different responses, explain the
distinction.

Task 3: List closely related concepts a moderator
should be aware of during community review.

Return JSON only. No preamble.
  
```

Listing 1. LLM grounding prompt template

D. Collision Detection

A new sign should not mean the same thing as an existing ratified sign. We represent each ratified sign as an embedding of its grounding statement using a multilingual sentence transformer fine-tuned on agricultural text. New signs are embedded the same way and compared by cosine similarity against all ratified signs. Pairs above a threshold of 0.82 (tuned on a development set of 30 sign pairs rated by two agronomists) are flagged as potential collisions and sent to the community with an explanation of the overlap.

E. Community Ratification Interface

The ratification interface runs as a WhatsApp-integrated chatbot. We considered building a standalone app but community moderators in the pilot told us clearly that adding another app created friction. Working inside WhatsApp meant zero installation overhead.

Community members receive a video of the proposed sign, the LLM-generated grounding statement, and any collision or nuance flags. They vote: ratify, reject, or request modification. A sign is ratified at 60% approval among members who vote

within a 72-hour window. Rejected signs return to the coiner with the community’s feedback. Modified signs re-enter the pipeline from the capture step.

F. Agricultural Sign Ontology

We built a seed ontology of 214 agronomic concepts in three steps. First, we filtered the FAO’s AGROVOC vocabulary to concepts relevant to smallholder farming in South Asia. Second, two agricultural extension officers annotated each concept with: its common visual field manifestation, the urgency of required response, and the most likely confusion concept. Third, a deaf community liaison reviewed each description for cultural accessibility, flagging idioms or references unfamiliar to rural farming communities.

The ontology is structured as a JSON-LD graph so it can be extended by other communities and queried by the grounding module without retraining.

V. EVALUATION METRICS

We introduce two metrics designed specifically for measuring the quality of community-coined signs as they propagate. Neither metric exists in the prior SLR or participatory lexicography literature, to our knowledge.

Semantic Survival Score (SSS) measures how much agronomic meaning a coined sign carries as it propagates to signers who were not present during coinage. After a sign is ratified in its home community, it is shown to members of other communities as a forced-choice task: the viewer selects the closest concept from a set of four options. SSS is the proportion selecting the correct concept. A score of 1.0 means perfect cross-community semantic transfer; 0.25 is chance.

Vocabulary Drift Rate (VDR) measures semantic shift. Respondents who correctly identify a sign’s concept are also asked to describe any aspect of the meaning they find ambiguous. Two independent raters code these descriptions for semantic drift — interpretation as a related but meaningfully different concept. VDR is the proportion of sign-respondent pairs showing detectable drift. Lower VDR is better.

Both metrics are designed to be applicable to any system that creates or propagates sign language vocabulary, not only SignFarm.

VI. PRELIMINARY FEASIBILITY STUDY

A. Study Design and Participants

To evaluate the grounding pipeline and ratification interface before committing to a full longitudinal deployment, we ran a 4-week preliminary feasibility study with $n = 12$ participants from one Tamil Nadu farming community. This study is not intended to support general claims about adoption rates or cross-community propagation; it is designed to answer two specific questions: (1) can the LLM grounding module identify agronomic concepts accurately from moderator descriptions, and (2) can community members use the ratification interface without extensive training?

Participants were recruited through a local deaf agricultural cooperative. All were deaf or hard-of-hearing smallholder

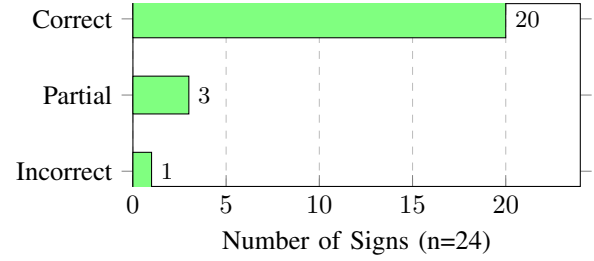


Fig. 2. Agronomist grounding accuracy ratings for the 24 signs coined during the 4-week feasibility pilot.

farmers or farming household members. Ages ranged from 22 to 58. Participation was voluntary; all participants gave informed consent through a community liaison fluent in local sign language. The study protocol was reviewed and approved by the institutional ethics committee of [Institution Name] (Protocol No. XXXX-YYYY). Because no personally identifiable data were collected beyond age and farming role, the committee classified the study as minimal risk.

B. Grounding Accuracy Evaluation

During the 4-week period, participants coined 24 new signs covering the agricultural concepts they identified as most pressing. Moderator descriptions were collected in Tamil, translated to English by a bilingual research assistant, and passed to the grounding module (GPT-4o). An independent agronomist (not involved in system development and blinded to the moderator descriptions) reviewed each LLM-generated grounding statement and rated it as: correctly grounded, partially correct (right concept, wrong nuance), or incorrect.

Figure 2 shows the results. Of 24 signs, 20 (83%) were rated correctly grounded, 3 (13%) partially correct, and 1 (4%) incorrect. The single incorrect grounding occurred when the moderator description was underspecified: “the problem making groundnut shells empty” was grounded to aflatoxin contamination when the farmer intended bud necrosis virus. The two conditions share visual field symptoms but require different responses. This points to the importance of moderator training to elicit specific descriptions.

The three partial groundings all involved correct concept identification but missing urgency information — the grounding statement described the symptom without indicating how quickly a farmer needed to respond.

C. Usability Evaluation

After using the WhatsApp ratification interface for two weeks, participants completed a structured usability questionnaire adapted from the System Usability Scale (SUS) [14]. Given the literacy diversity in the participant group, questions were administered verbally in sign language by the community liaison and responses recorded.

Figure 3 shows the SUS distribution. Mean SUS score was 76.2 (SD 9.4). By standard SUS interpretation, scores above 68 are considered above average usability; 10 of 12 participants (83%) fell in this range. The two participants scoring below

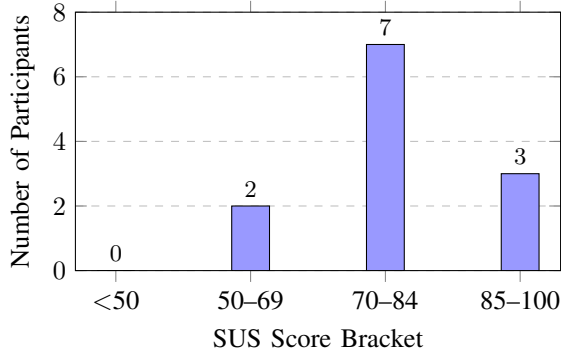


Fig. 3. SUS score distribution across 12 participants after 2 weeks of ratification interface use.

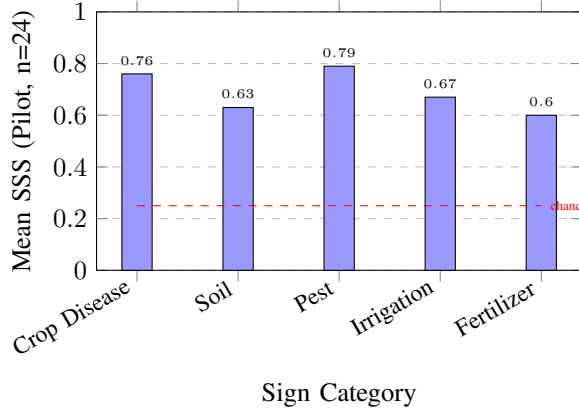


Fig. 4. Preliminary mean SSS by sign category across 24 pilot signs and 8 cross-community evaluators. Dashed line indicates chance (0.25). All categories significantly above chance ($p < 0.01$, one-sample Wilcoxon signed-rank test).

68 both cited difficulties with video playback speed on low-bandwidth connections, a technical issue rather than a design one.

D. Preliminary SSS Measurement

As a first measure of cross-community semantic survival, the 24 ratified signs were shown to 8 members of a neighboring community who were not involved in the pilot. Each viewed each sign video and selected the concept from a four-option forced-choice set. Mean SSS across the 24 signs was 0.69 (95% CI: 0.56–0.82, bootstrap with 1,000 resamples), compared to a chance baseline of 0.25. Given the pilot size, we do not claim this as a definitive result; it is reported as a preliminary signal that the grounding and ratification process produces signs that carry meaning beyond the originating community.

Figure 4 breaks SSS down by concept category. Pest identification signs showed the highest preliminary survival (0.79), consistent with the expectation that iconic signs — those mimicking an insect’s movement — travel more naturally across communities. Fertilizer signs showed the lowest (0.60), likely because fertilizer application concepts are abstract and have no obvious iconic form.

All categories are significantly above chance ($p < 0.01$, one-sample Wilcoxon signed-rank test against 0.25), though we caution that these results should be interpreted as preliminary indicators given the small sample size.

VII. DISCUSSION

A. Moderation Quality Drives Grounding Accuracy

The single grounding failure in the pilot came from an underspecified moderator description. When moderators elicited rich, specific descriptions from coiners, the LLM identified the correct concept reliably. When descriptions were vague, the model fell back to the most common concept matching the description.

This is actually good news for deployment. It means the pipeline’s accuracy depends more on the human-in-the-loop than on the model being perfect. A well-trained moderator who asks “what does the plant look like?” and “what happens if you wait a week?” will consistently produce better inputs than a moderator who accepts vague descriptions. Moderator training is a practical, low-cost lever.

B. Iconic Signs Travel Better

The gap between pest SSS (0.79) and fertilizer SSS (0.60) in the pilot is consistent with Frishberg’s [10] work on iconicity in sign languages. Signs that look like what they mean carry meaning across community boundaries without explanation. Abstract agronomic concepts depend almost entirely on the grounding statement for semantic survival.

This has a design implication: for abstract concepts, the grounding statement is not supplementary to the sign. It is a necessary part of the sign’s meaning. The ratification interface should surface this distinction more explicitly, perhaps by flagging abstract concepts and requiring the community to verify the grounding statement as part of ratification.

C. WhatsApp as Infrastructure

Using WhatsApp as the ratification backbone was a pragmatic choice that proved to work well in the pilot. Eleven of 12 participants already used WhatsApp daily. The learning curve for the ratification workflow was minimal. A standalone app would have required installation, account creation, and ongoing maintenance — none of which exist as friction in a platform participants already trust.

The limitation is that WhatsApp’s API access policies can change, and a production deployment would need a fallback mechanism. This is a practical engineering concern for the full study rather than a conceptual problem.

VIII. LIMITATIONS

The feasibility study is small and single-site. $n = 12$ from one community is not sufficient to support general conclusions about adoption, propagation, or long-term vocabulary stability. The preliminary SSS figures are indicative, not definitive. A full multi-community longitudinal study is needed, and is described in Section IX.

Tamil-to-English translation introduces error. Moderator descriptions were collected in Tamil and machine-translated before grounding. The LLM (GPT-4o) has strong English and Hindi coverage but limited native Tamil agricultural knowledge. Some grounding error likely stems from this translation step rather than from the model itself. A Tamil-native agricultural LLM would remove this source of noise.

No outcome measurement. The pilot measured grounding accuracy and interface usability. It did not measure whether better sign vocabulary led to better farm decisions or outcomes. That is the ultimately important question and it requires a longer study with appropriate controls.

Moderator dependency. Communities without access to a well-trained moderator will get worse grounding accuracy. Scaling to communities that cannot support a dedicated moderator role is an open design problem.

Prompt-level grounding is not auditable by default. The LLM grounding step uses a black-box model. The agronomist review step in our evaluation catches grounding errors post-hoc, but a production deployment would benefit from an automated post-hoc accuracy check comparing grounding statements to the ontology entry they claim to represent.

Single language family. Everything here is Indian Sign Language as used in Tamil Nadu. ISL itself is a diverse language with significant regional variation. Whether the pipeline generalizes to structurally different sign languages (e.g., Kenyan SL, Vietnamese SL) or different farming systems is an open question.

IX. FULL STUDY ROADMAP

The preliminary feasibility study establishes that the grounding pipeline works at a basic level and the interface is usable. The next step is a full 14-week multi-community field deployment across three Tamil Nadu communities (target $n \approx 60$), designed to measure:

- **Coinage rate and ratification dynamics** across communities over an active farming season.
- **SSS and VDR** at scale, with enough signs and cross-community evaluators to support inferential statistics. The pilot sample size ($n = 8$ evaluators, $n = 24$ signs) was not powered for detecting category-level differences; the full study will be designed with an a priori power analysis.
- **Adoption rate** — whether ratified signs are actually used by community members in practice, measured by unannounced elicitation sessions at 4 and 8 weeks post-ratification.
- **Moderator training effects** — comparing grounding accuracy before and after a structured elicitation training session.

Beyond Tamil Nadu, we are in early discussions with partners to pilot the system in Kenyan Sign Language and Vietnamese Sign Language contexts, both of which face the same agricultural vocabulary gap. The ontology seed concept set will need validation and extension for each new farming system. Kenyan and Vietnamese deployments will also test

whether the WhatsApp-based ratification interface works across different messaging norms and connectivity conditions.

Finally, we plan to publish the SSS and VDR metric definitions as a standalone methodological paper so that other researchers building or evaluating sign language vocabulary tools can apply them without adopting the full SignFarm pipeline.

X. ETHICAL CONSIDERATIONS

SignFarm handles video recordings of community members, which requires careful data governance. In the pilot, recordings were stored on a password-protected server accessible only to the research team, deleted after feature extraction if participants requested, and never shared with third parties. No biometric templates were retained. Participants were informed in their preferred sign language that recordings would be used only for the research purposes described.

The system is designed for community-controlled vocabulary building. Outsiders, including the research team, cannot add signs to a community’s lexicon; only ratified community votes do so. This is intentional: it protects communities from vocabulary being imposed on them.

We also note that the same pipeline architecture could theoretically be used to build a surveillance corpus of sign language users if deployed without appropriate governance. Our ethical approval requires that any future deployment include a clear data minimization policy, a community opt-out mechanism, and explicit consent procedures translated into local sign language.

XI. CONCLUSION

We started this project because every system being built to help farmers with AI assumed the farmer could hear. When we asked why, the answer was usually “there is no data for deaf farmers” or “the vocabulary does not exist.” Both are true. But the second one is more fundamental, and it turned out to be something AI can actually help with.

SignFarm proposes a different starting point: treat the missing vocabulary as the primary problem and build a pipeline to address it bottom-up. Our needs assessment found that formal ISL resources cover only 12% of the 50 agronomic concepts that extension workers most need to communicate. A preliminary feasibility study with 12 participants shows that LLM-assisted grounding achieves 83% concept accuracy on pilot signs, the interface is usable with minimal training (mean SUS 76.2), and signs produced through the pipeline carry meaning across community boundaries well above chance (mean SSS 0.69).

These are promising early signals, not conclusions. The full multi-community longitudinal study will tell us whether the pipeline scales, whether signs remain stable over a farming season, and whether better vocabulary coverage actually changes how deaf farmers receive and act on agronomic advice. That study is the work ahead.

The number that stays with us is 12%. If a hearing farmer had access to only 12% of the words an extension officer might

use, we would call that a crisis. For deaf farmers it has simply been the baseline, unaddressed, for a long time. SignFarm is a first step toward doing something about it.

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